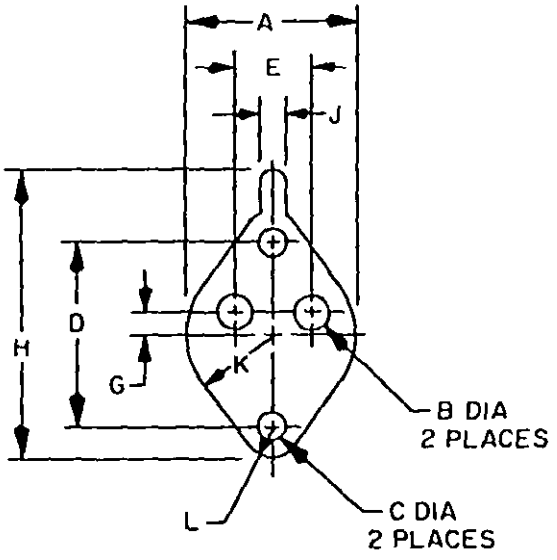
MILITARY SPECIFICATION SHEET

HEAT SINKS-INSULATORS, CLASS 1, 4, 6, AND 7 (WAFER TYPE)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-H-87111.

Part or identifying numbers (PINs) M87111/01-4Y15 through M87111/01-4Y33 are inactive for new design as of 9 May 1989.

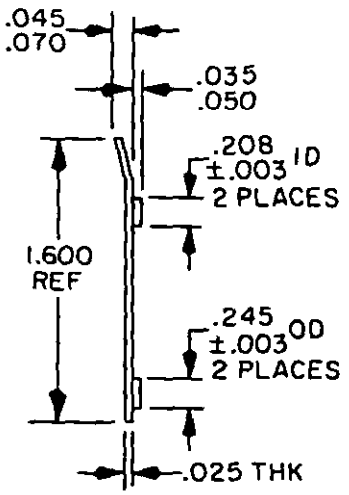


TOP VIEW
CONFIGURATION A - Dash numbers 1H01 through 1H08

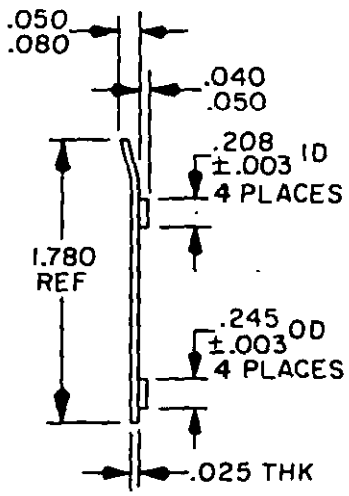
FIGURE 1. Heat sinks (wafer type).

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

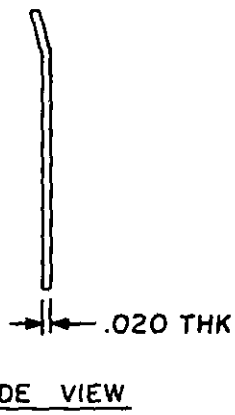
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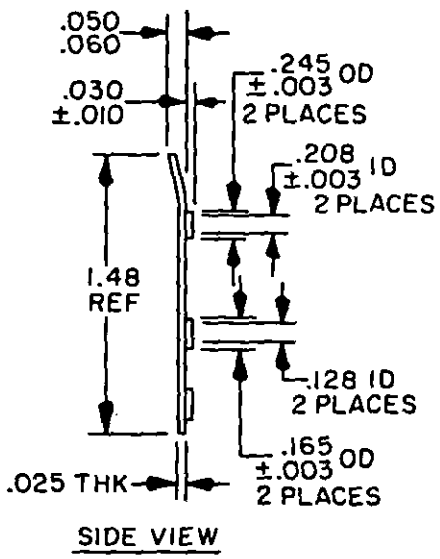
Dash number 1H03



Dash number 1H04



Dash numbers 1H01, 1H02, 1H05, 1H06, and 1H07



Dash number 1H08

CONFIGURATION A - Dash numbers 1H01 through 1H08

FIGURE 1. Heat sinks (wafer type) - Continued.

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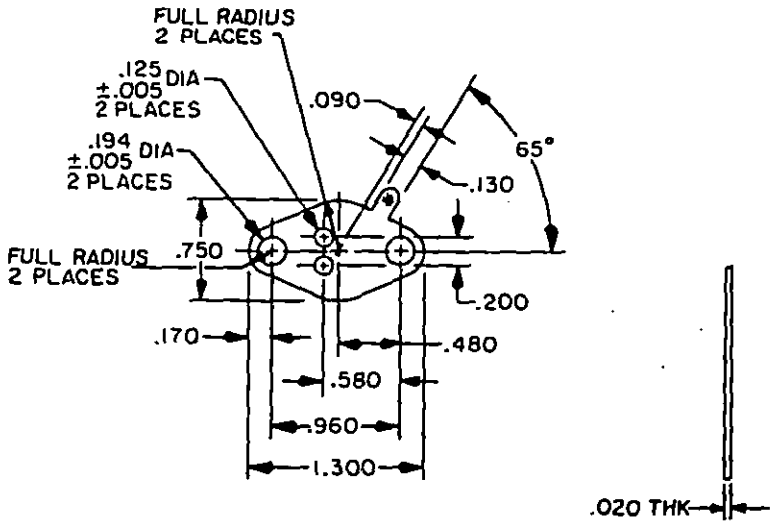
* PIN M87111/01-	Top view dimensions										JEDEC case size
	A	B	C	D	E	G	H	J	K	L	
1H01	1.063 (27.00)	.090 (2.29)	.156 (3.96)	1.187 (30.15)	.430 (10.92)	.070 (1.78)	1.793 (45.54)	.187 (4.75)	.532 (13.51)	.203 (5.16)	TO-3
1H02	1.063 (27.00)	.220 (5.59)	.190 (4.83)	1.187 (30.15)	.430 (10.92)	.070 (1.78)	1.793 (45.54)	.187 (4.75)	.532 (13.51)	.203 (5.16)	TO-3
1H03	1.038 (26.37)	.215 (5.46)	.208 (5.28)	1.187 (30.15)	.430 (10.92)	.070 (1.78)	1.600 (40.64)	.190 (4.83)	.519 (13.18)	.180 (4.57)	TO-3
1H04	1.170 (29.72)	.208 (5.28)	.208 (5.28)	1.187 (30.15)	.430 (10.92)	.070 (1.78)	1.780 (45.21)	.190 (4.83)	.585 (14.86)	.240 (6.10)	TO-3
1H05	1.000 (25.40)	.202 (5.13)	.156 (3.96)	1.187 (30.15)	.437 (11.10)	.070 (1.78)	1.700 (43.18)	---	.500 (12.70)	.198 (5.03)	TO-3
1H06	1.125 (28.58)	.218 (5.54)	.196 (4.98)	1.187 (30.15)	.430 (10.92)	.070 (1.78)	1.950 (49.53)	.250 (6.35)	.563 (14.30)	.206 (5.23)	TO-3
1H07	.710 (18.03)	.062 (1.57)	.140 (3.56)	.960 (24.38)	.200 (5.08)	.100 (2.54)	1.440 (36.58)	.180 (4.57)	.355 (9.02)	.150 (3.81)	TO-66
1H08	.820 (20.83)	.128 (3.25)	.208 (5.28)	.960 (24.38)	.200 (5.08)	.100 (2.54)	1.480 (37.59)	.188 (4.78)	.410 (10.41)	.210 (5.33)	TO-66

* Part or identifying number (PIN).

CONFIGURATION A - Continued.

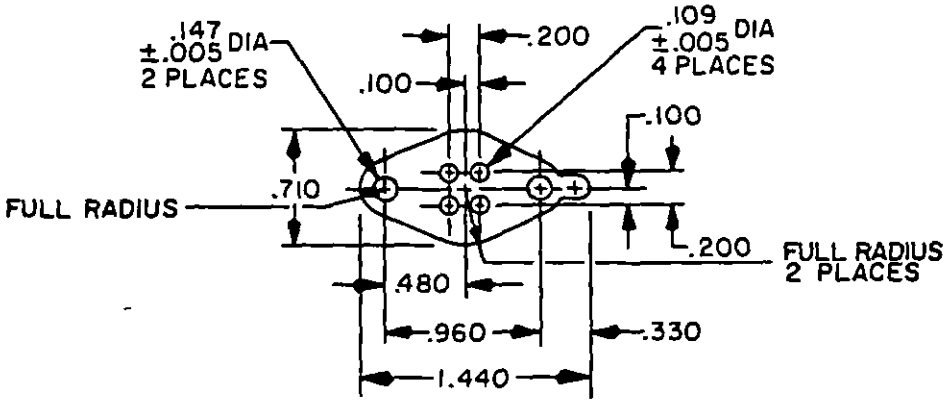
FIGURE 1. Heat sinks (wafer type) - Continued.

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Check the source to verify that this is the current version before use.



CONFIGURATION B
Dash number 1H09
(TO-66)

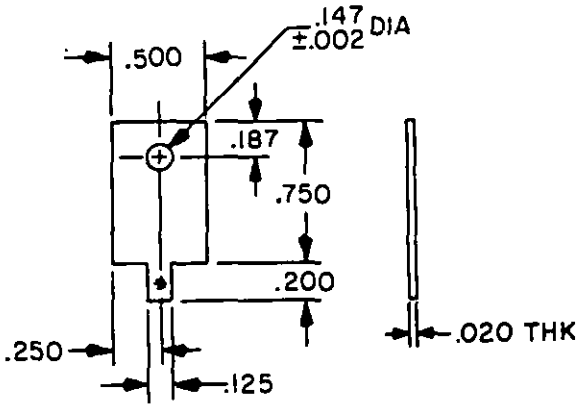
SIDE VIEW
Dash numbers 1H09 and 1H10



CONFIGURATION C
Dash number 1H10
(TO-66)

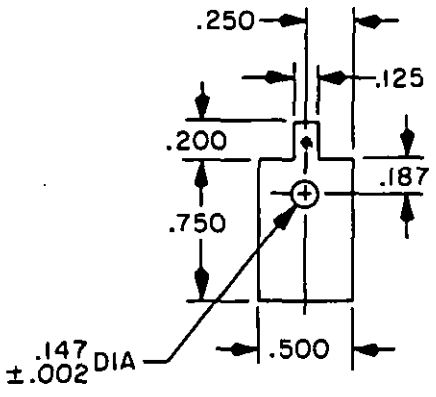
FIGURE 1. Heat sinks (wafer type) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:48Z
Check the source to verify that this is the current version before use.

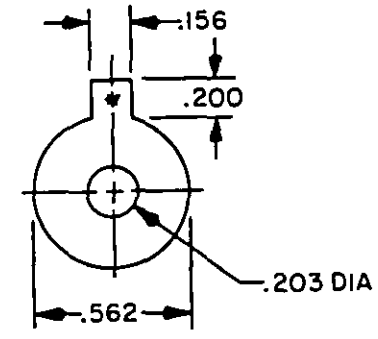


CONFIGURATION D
Dash number 1HI1

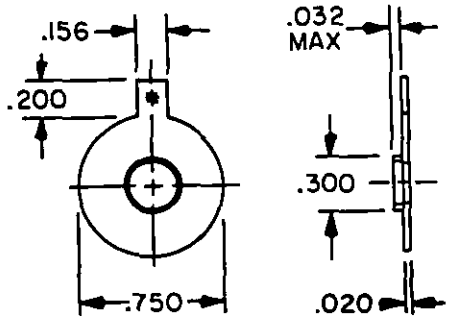
SIDE VIEW
Dash numbers 1H11 and 1H12



CONFIGURATION E
Dash number 1HI2



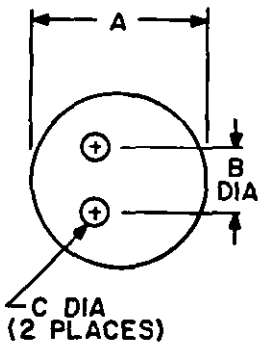
CONFIGURATION F
Dash number 1HI3



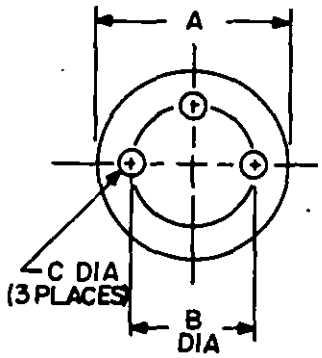
CONFIGURATION G
Dash number 1HI4

FIGURE 1. Heat sinks (wafer type) - Continued.

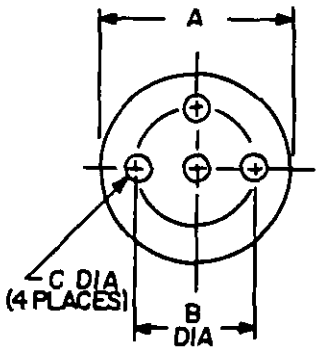
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:48Z
Check the source to verify that this is the current version before use.



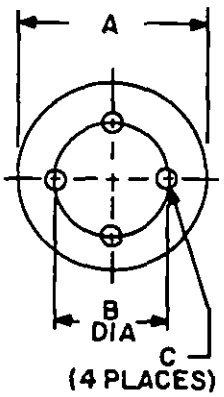
CONFIGURATION H



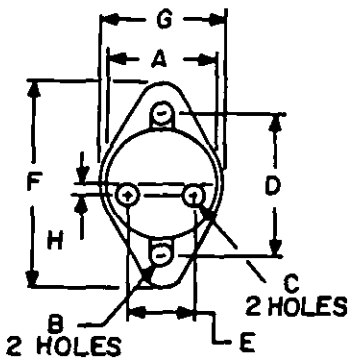
CONFIGURATION J



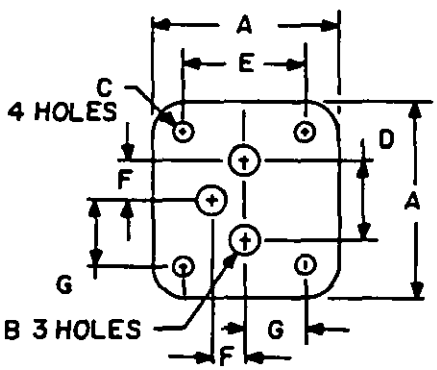
CONFIGURATION K



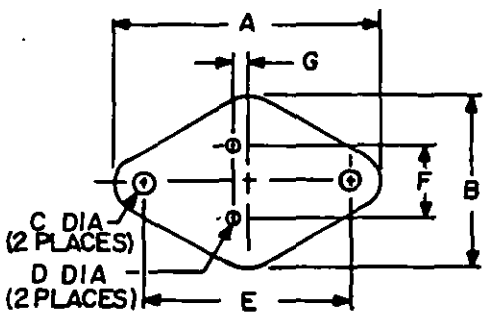
CONFIGURATION L



CONFIGURATION M



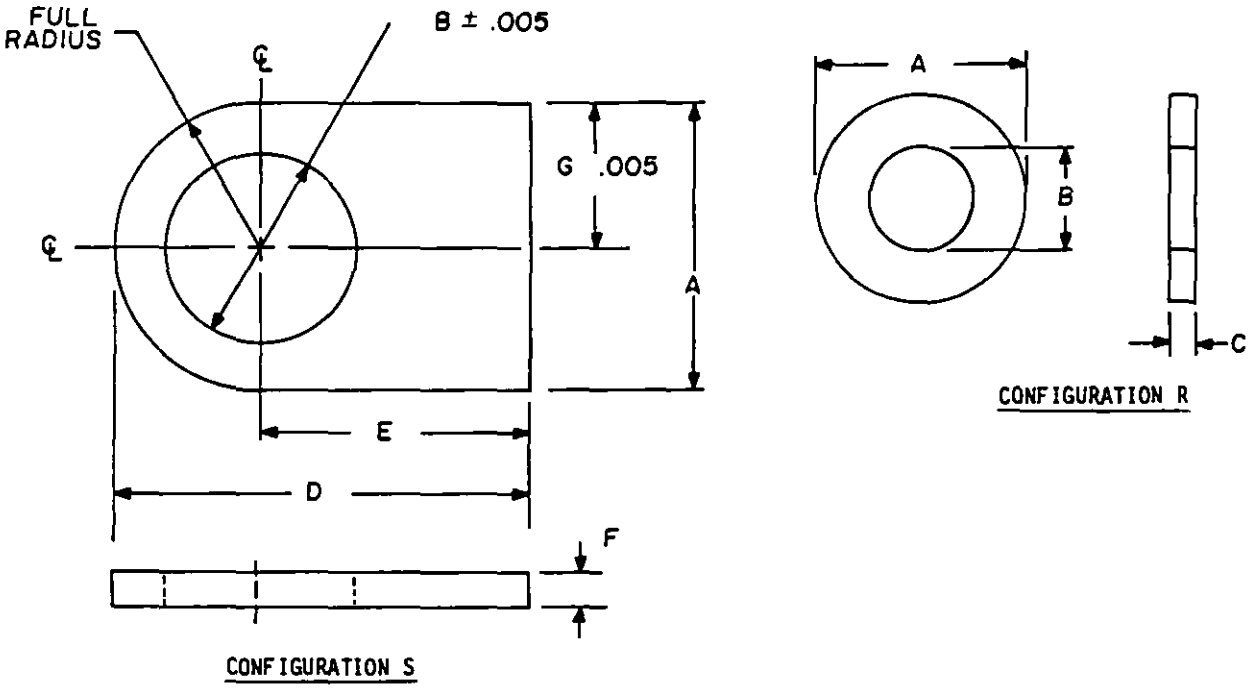
CONFIGURATION N



CONFIGURATION P

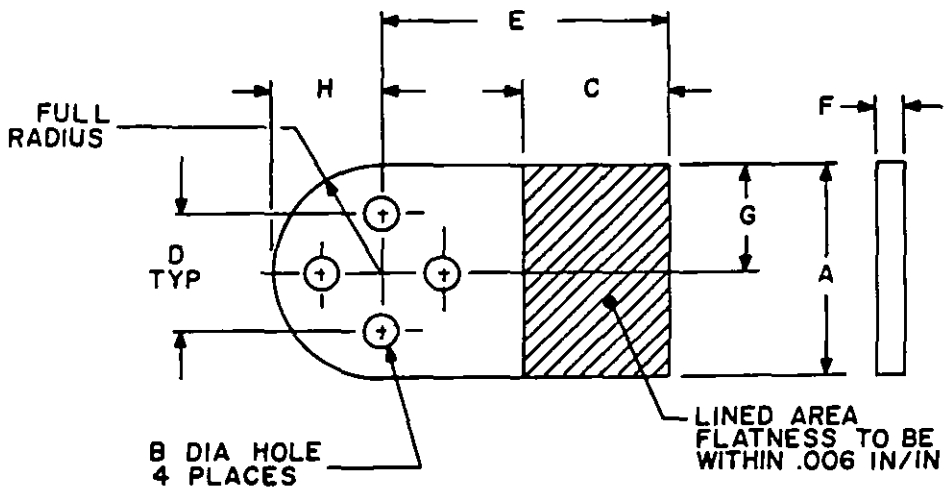
FIGURE 1. Heat sinks (wafer type) - Continued.

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CONFIGURATION S

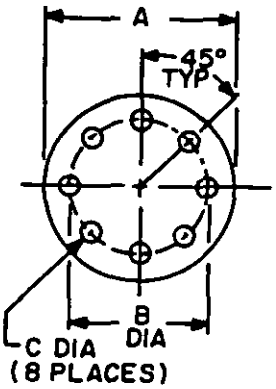
CONFIGURATION R



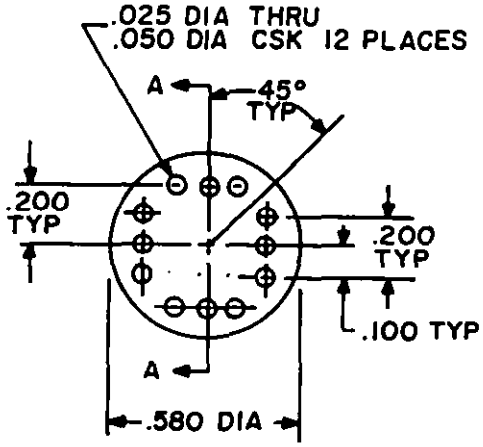
CONFIGURATION T

FIGURE 1. Heat sinks (wafer type) - Continued.

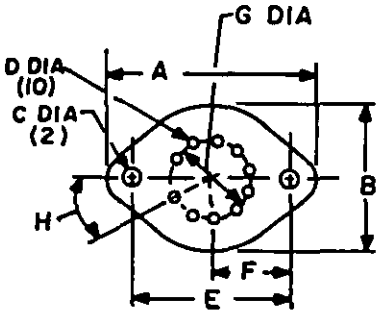
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:48Z
Check the source to verify that this is the current version before use.



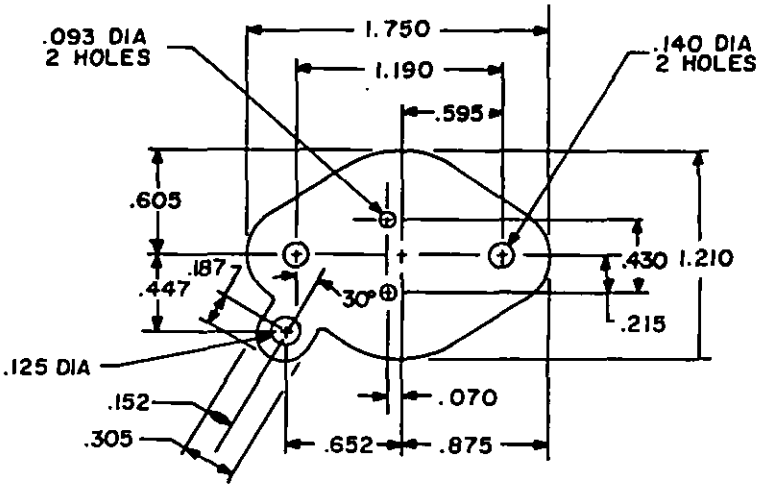
CONFIGURATION V



CONFIGURATION W
Dash number *Y70



CONFIGURATION X



CONFIGURATION Z
Dash number *Y71
(TO-3)

FIGURE 1. Heat sinks (wafer type) - Continued.

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PIN M87111/01- (see note 7)	JEDEC case size	Config-uration	Dimensions (see note 8)							
			A	B	C	D	E	F	G	H
4Y15	TO-3	H	.980 (24.89)	.435 (11.05)	.165 (4.19)	---	---	---	---	---
4Y16	TO-18	J	.220 (5.59)	.100 (2.54)	.040 (1.02)	---	---	---	---	---
4Y17 (See note 9)	TO-5	J	.360 (9.14)	.200 (5.08)	.040 (1.02)	---	---	---	---	---
4Y18	TO-8	J	.625 (15.88)	.282 (7.16)	.050 (1.27)	---	---	---	---	---
4Y19	TO-36	K	1.063 (27.00)	.690 (17.53)	.200 (5.08)	---	---	---	---	---
4Y20	TO-3	L	.437 (11.10)	.200 (5.08)	.054 (1.37)	---	---	---	---	---
4Y21	TO-3	M	1.000 (25.40)	.140 (3.56)	.165 (4.19)	1.187 (30.15)	.435 (11.05)	1.563 (39.70)	1.125 (28.58)	---
4Y22	TO-53	N	.860 (21.84)	.171 (4.34)	.109 (2.77)	.400 (10.16)	.680 (17.27)	.200 (5.08)	.340 (8.64)	---
4Y23	TO-53	N	1.100 (27.94)	.174 (4.42)	.116 (2.95)	.400 (10.16)	.680 (17.27)	.200 (5.08)	.340 (8.64)	---

FIGURE 1. Heat sinks (wafer type) - Continued.

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Check the source to verify that this is the current version before use.

PIN M87111/01- (see note 7)	JEDEC case size	Config-uration	Dimensions (see note 8)							
			A	B	C	D	E	F	G	H
4Y24	TO-66	P	1.250 (31.75)	.700 (17.78)	.147 (3.73)	.130 (3.30)	.960 (24.38)	.200 (5.08)	.100 (2.54)	---
4Y25	TO-3	P	1.563 (39.70)	1.050 (26.67)	.145 (3.68)	.145 (3.68)	1.187 (30.15)	.430 (10.92)	.072 (1.83)	---
4Y26	Stud type	R	.470 (11.94)	.235 (5.97)	.040 (1.02)	---	---	---	---	---
4Y27	Stud type	R	.510 (12.95)	.200 (5.08)	.040 (1.02)	---	---	---	---	---
4Y28	Stud type	R	.800 (20.32)	.260 (6.60)	.060 (1.52)	---	---	---	---	---
4Y29	Stud type	R	.800 (20.32)	.320 (8.13)	.060 (1.52)	---	---	---	---	---
4Y30	Stud type	R	.900 (22.86)	.320 (8.13)	.060 (1.52)	---	---	---	---	---
4Y31		S	.500 (12.70)	.350 (8.89)	---	.730 (18.54)	.480 (12.19)	.050 (1.27)	.250 (6.35)	---
4Y32		T	.380 (9.65)	.050 (1.27)	.250 (6.35)	.200 (5.08)	.500 (12.70)	.040 (1.02)	.187 (4.75)	.187 (4.75)
4Y33		T	.250 (6.35)	.027 (0.69)	.250 (6.35)	.100 (2.54)	.425 (10.80)	.040 (1.02)	.125 (3.18)	.125 (3.18)
1H34 (see note 10)		P	1.030 (26.16)	.220 (5.59)	.190 (4.83)	1.187 (30.15)	.430 (10.92)	1.547 (39.29)	---	.070 (1.78)

FIGURE 1. Heat sinks (wafer type) - Continued.

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PIN M87111/01- (see notes 7 and 12)	JEDEC case size	Config-uration	Dimensions (see notes 10 and 11)							
			A	B	C	D	E	F	G	H
*Y35	TO-3	H	.980 (24.89)	.435 (11.05)	.165 (4.19)	---	---	---	---	---
*Y36	TO-18	J	.220 (5.59)	.100 (2.54)	.040 (1.02)	---	---	---	---	---
*Y37	TO-5 TO-39	J	.360 (9.14)	.200 (5.08)	.040 (1.02)	---	---	---	---	---
*Y38	TO-8	J	.625 (15.88)	.282 (7.16)	.050 (1.27)	---	---	---	---	---
*Y39	TO-36	K	1.063 (27.00)	.690 (17.53)	.200 (5.08)	---	---	---	---	---
*Y40	TO-5	L	.360 (9.14)	.200 (5.08)	.054 (1.37)	---	---	---	---	---
*Y41	--	L	.437 (11.10)	.200 (5.08)	.054 (1.37)					
*Y42	TO-53	N	.860 (21.84)	.171 (4.34)	.109 (2.77)	.400 (10.16)	.680 (17.27)	.200 (5.08)	.340 (8.64)	.200 (5.08)
*Y43	TO-53	N	1.100 (27.94)	.174 (4.42)	.116 (2.95)	.400 (10.16)	.680 (17.27)	.200 (5.08)	.340 (8.64)	.200 (5.08)

FIGURE 1. Heat sinks (wafer type) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:48Z
Check the source to verify that this is the current version before use.

PIN M87111/01- (see notes 7 and 12)	JEDEC case size	Config-uration	Dimensions (see notes 10 and 11)							
			A	B	C	D	E	F	G	H
*Y44	TO-66	P	1.250 (31.75)	.700 (17.78)	.147 (3.73)	.130 (3.30)	.960 (24.38)	.200 (5.08)	.100 (2.54)	---
*Y45	TO-66	P	1.312 (33.32)	.762 (19.35)	.140 (3.56)	.062 (1.57)	.960 (24.38)	.200 (5.08)	.100 (2.54)	---
*Y46	TO-66	P	1.375 (34.93)	.825 (20.96)	.140 (3.56)	.062 (1.57)	.960 (24.38)	.200 (5.08)	.100 (2.54)	---
*Y47	TO-66	P	1.440 (36.58)	1.000 (25.40)	.140 (3.56)	.075 (1.91)	.960 (24.38)	.200 (5.08)	.100 (2.54)	---
*Y48	TO-3	P	1.563 (39.70)	1.050 (26.67)	.140 (3.56)	.080 (2.03)	1.187 (30.15)	.430 (10.92)	.072 (1.83)	---
*Y49	TO-3	P	1.563 (39.70)	1.050 (26.67)	.145 (3.68)	.145 (3.68)	1.187 (30.15)	.430 (10.92)	.072 (1.83)	---
*Y50	TO-3	P	1.563 (39.70)	1.125 (28.58)	.140 (3.56)	.165 (4.19)	1.187 (30.15)	.435 (11.05)	.070 (1.78)	---
*Y51	TO-3	P	1.593 (40.46)	1.100 (27.94)	.156 (3.96)	.062 (1.57)	1.187 (30.15)	.430 (10.92)	.072 (1.83)	---
*Y52	TO-3	P	1.650 (41.91)	1.065 (27.05)	.140 (3.56)	.046 (1.17)	1.187 (30.15)	.430 (10.92)	.072 (1.83)	---
*Y53	TO-3	P	1.650 (41.91)	1.140 (28.96)	.122 (3.10)	.062 (1.57)	1.187 (30.15)	.430 (10.92)	.072 (1.83)	---
*Y54	TO-3	P	1.650 (41.91)	1.140 (28.96)	.140 (3.56)	.093 (2.36)	1.187 (30.15)	.430 (10.92)	.072 (1.83)	---
*Y55	TO-3	P	1.650 (41.91)	1.140 (28.96)	.165 (4.19)	.062 (1.57)	1.187 (30.15)	.430 (10.92)	.072 (1.83)	---
*Y56	TO-3	P	1.700 (43.18)	1.187 (30.15)	.156 (3.96)	.062 (1.57)	1.187 (30.15)	.430 (10.92)	.072 (1.83)	---

FIGURE 1. Heat sinks (wafer type) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T11:48Z
Check the source to verify that this is the current version before use.

PIN M87111/01- (see notes 7 and 12)	JEDEC case size	Config-uration	Dimensions (see notes 10 and 11)							
			A	B	C	D	E	F	G	H
*Y57	TO-3	P	1.780 (45.21)	1.250 (31.75)	.140 (3.56)	.093 (2.36)	1.187 (30.15)	.430 (10.92)	.072 (1.83)	
*Y58	TO-3	X	1.650 (41.91)	1.140 (28.96)	.165 (4.19)	.040 (1.02)	1.187 (30.15)	.593 (15.06)	.500 (12.70)	32.70 (830.6)
*Y59	DO-4	R	.510 (12.95)	.200 (5.08)	---	---	---	---	---	---
*Y60	DO-4	R	.625 (15.88)	.200 (5.08)	---	---	---	---	---	---
*Y61	DO-5	R	.800 (20.32)	.260 (6.60)	---	---	---	---	---	---
*Y62	DO-5	R	1.000 (25.40)	.260 (6.60)	---	---	---	---	---	---
*Y63	--	R	.437 (11.10)	.235 (5.97)	---	---	---	---	---	---
*Y64	--	R	.800 (20.32)	.320 (8.13)	---	---	---	---	---	---
*Y65	--	R	.900 (22.86)	.320 (8.13)	---	---	---	---	---	---
*Y66	--	S	.500 (12.70)	.350 (8.89)	---	.730 (18.54)	.480 (12.19)	.050 (1.27)	.250 (6.35)	---
*Y67	--	T	.380 (9.65)	.050 (1.27)	.250 (6.35)	.200 (5.08)	.500 (12.70)	.040 (1.02)	.187 (4.75)	.187 (4.75)
*Y68	--	T	.250 (6.35)	.027 (0.69)	.250 (6.35)	.100 (2.54)	.425 (10.80)	.040 (1.02)	.125 (3.18)	.125 (3.18)
*Y69	TO-99	V	.390 (9.91)	.200 (5.08)	.040 (1.02)					

FIGURE 1. Heat sinks (wafer type) - Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.045	1.14	.128	3.25	.208	5.28	.750	19.05
.003	0.08	.050	1.27	.130	3.30	.245	6.22	.960	24.38
.005	0.13	.060	1.52	.147	3.73	.250	6.35	1.300	33.02
.010	0.25	.070	1.78	.156	3.96	.300	7.62	1.440	36.58
.020	0.51	.080	2.03	.165	4.19	.330	8.38	1.480	37.59
.025	0.64	.090	2.29	.170	4.32	.480	12.19	1.600	40.64
.030	0.76	.100	2.54	.187	4.75	.500	12.70	1.780	45.21
.032	0.81	.109	2.77	.194	4.93	.562	14.27		
.035	0.89	.125	3.18	.200	5.08	.580	14.73		
.040	1.02			.203	5.16	.710	18.03		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Metric equivalents are in parentheses.
4. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for decimals and ±2° for angles. A .010 (0.25 mm) radius on corners and a 1° draft angle on sides are permitted.
5. Unless otherwise specified, thickness is .020 (0.51 mm).
6. Unless otherwise specified, thickness tolerances for class 1 parts is ±.004 (0.10 mm).
7. Unless otherwise specified, tolerance is ±.015 (0.38 mm) on width, length, hole diameter, and hole location.
8. Thickness shall be .062 (1.57 mm) unless otherwise specified.
9. Thickness shall be .030 (0.76 mm).
10. Thickness shall be .020 (0.51 mm).
11. Thickness shall be .005 (0.13 mm) to .080 (2.03 mm), depending upon material type and application.
12. Complete PIN will include a class number (6 or 7) in place of the asterisk to denote material.
13. Notes 4 through 6 pertain to configurations A through G.
14. Notes 7 through 12 pertain to configurations H through Z.

FIGURE 1. Heat sinks (wafer type) - Continued.

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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Material options:
Aluminum alloy (class 1).
Beryllium oxide (class 4).
Boron nitride-filled silicone rubber (class 6).
Alumina-filled silicone rubber (class 7).

CAUTION: Class 4 devices contain beryllium oxide ceramics. Heat sink-insulators of this type are not hazardous in their finished state; however, grinding, sandblasting, machining, sectioning, or similar operations should not be performed on these parts unless proper safety precautions are observed. These operations may produce beryllium dust or fumes which, if inhaled, could cause serious pulmonary disease. Inhalation of beryllium particles in excess of OSHA occupational exposure standards should be avoided. Class 4 parts may be recycled or disposed of according to the manufacturer's instructions.

Finish options:
Hard anodize (finish H).
No finish (finish Y).

JEDEC case sizes: See figure 1.

Mounting surface flatness: Mounting surfaces of class 4 parts shall be flat within .002 inch (0.05 mm) per inch of Total Indicator Reading (TIR).

Surface roughness: Mounting surface shall not exceed 25 microinches rms.

Thermal resistance:

a. Class 1: 0.4°C/W maximum.

b. Class 4: 1.5°C/W maximum.

c. Class 6: 0.4°C/W maximum.

d. Class 7: 0.6°C/W maximum.

PIN: M87111/01- (plus dash number from figure 1). Complete PIN example: M87111/01-6Y48.

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TABLE I. Supersession data.

Superseding PIN M87111/01- 1/	Superseded PIN in accordance with MIL-H-87111/1B 25 September 1985	Superseding PIN M87111/01- 1/	Superseded PIN in accordance with MIL-H-87111/1B 25 September 1985
*Y35	M87111/01-4Y15	*Y49	M87111/01-4Y25
*Y36	M87111/01-4Y16	*Y63	M87111/01-4Y26
*Y37	M87111/01-4Y17	*Y59	M87111/01-4Y27
*Y38	M87111/01-4Y18	*Y61	M87111/01-4Y28
*Y39	M87111/01-4Y19	*Y64	M87111/01-4Y29
*Y41	M87111/01-4Y20	*Y65	M87111/01-4Y30
*Y50	M87111/01-4Y21	*Y66	M87111/01-4Y31
*Y42	M87111/01-4Y22	*Y67	M87111/01-4Y32
*Y43	M87111/01-4Y23	*Y68	M87111/01-4Y33
*Y44	M87111/01-4Y24		

1/ Superseding PINs are effective 9 May 1989.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - TD
Air Force - 11, 19, 99
DLA - ES

User activities:
Navy - AS, CG, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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